

# Expression and Purification of Tobacco Etch Virus Protease

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## Abstract

TEV protease is an essential tool for removing affinity tags from recombinant proteins. Affinity tags are very helpful for detecting and purifying target proteins. However, these affinity tags can alter the composition of the protein. The sequence-specific TEV protease cleaves recombinant proteins of their histidine tags. Other proteases cut the recombinant proteins in areas not desired. This is why TEV protease has become a valuable tool in the laboratory for controlling protein cleavage. In this study, a bacterial plasmid containing the gene to produce TEV protease was obtained and transformed into an expression strain. The protein was expressed at a small scale and a large scale. TEV protease was then purified from the large-scale expression using affinity chromatography. The protein was desalted by dialysis and stored in 20% glycerol at -80°C for long-term storage.

## Introduction

The use of affinity tags has become a widespread tool in the biochemical world. These tags can be used for detection, protection, purification, and many more applications on proteins. However, some of these tags can affect the structure and function of proteins. Given that, it is best to remove these tags [1-2]. Proteases are enzymes that break down proteins. A class of viral proteases, such as tobacco etch virus N1a proteinase (TEV protease), is efficient for removal of these tags due to their high sequence specificity. There are problems with other proteases due to uncontrolled cleavage of the protein, which might cut the protein at unwanted sites. TEV protease will cut recombinant proteins specifically at the TEV site to remove affinity tags. TEV protease recognizes the peptide sequence Glu-X-X-Tyr-X-Gln-Gly/Ser, where X can be any residue, and it cleaves between Gln and Gly or Ser [3].

There are some problems with the production of TEV protease, for example, its low solubility [2]. A common approach is to use a fusion protein, which can increase its solubility. TEV protease is produced with a fusion protein called maltose-binding protein (MBP) on the C-terminus. MBP allows the protease to accumulate in a soluble, active form and helps prevent it from degrading. The fusion protein cleaves itself *in vivo* and is purified using affinity chromatography, leaving soluble TEV protease with a polyhistidine tag on the N-terminus.

This study demonstrates the production of MBP-His7-TEV from an *Escherichia Coli* expression vector. Production was carried out at a small scale of 100mL of LB medium and a large scale of 1L of LB medium. After reaching an OD<sub>600</sub> of about 0.5, growth media were induced with IPTG. Using affinity chromatography, His7-TEV was purified in a

soluble form. This protease is now used to cleave affinity tags from recombinant proteins in Karen Draths laboratory at Michigan State University.

## Methods

### *Expression Vector*

The expression vector pMHT $\Delta$ 238 containing the gene for TEV protease was obtained from Heedeok Hong at Michigan State University. The MHT coding region produces the fusion protein MBP-His<sub>7</sub>-TEV, where the TEV cleavage site is between the MBP and the histidine tag. pMHT $\Delta$ 238 is kanamycin-resistant and has a T5 promoter. It was transformed into electrocompetent BL21(DE3) C+ RP cells, which are ideal for protein expression with vectors with non- T7 promoters.

### *Small Scale Production*

Colonies from the transformation of pMHT $\Delta$ 238 into BL21(DE3) C+ RP cells were inoculated into 5mL of LB growth medium with Kanamycin (50mg/mL). This starting inoculum was incubated overnight at 37°C and reached an OD<sub>600</sub>  $\approx$  7. A fraction of the overnight inoculum was added to 100mL of expression medium containing LB and Kanamycin (50mg/mL). Culture was induced with IPTG when OD<sub>600</sub>  $\approx$  0.5 and was treated for 5 hours at 30°C. Small cultures were harvested by centrifugation at 6000 rpm for 5 minutes. Cells were lysed by enzymatic lysis. The lysis buffer was 20mM sodium phosphate (pH 7.4), 500mM sodium chloride, and 20mM imidazole. Lysozyme (1mg/mL) was added, followed by Triton X-100 (1%) and DNase. SDS-PAGE of the supernatant confirmed His<sub>7</sub>-TEV was produced and self-cleaved from MBP. Figure 1 represents the four small culture flasks. Control and

treatment flasks all grew under the same conditions. The last lane was the flask that contained plasmid pMHT $\Delta$ 238 and was induced with IPTG. This confirmed that the protein expression worked and TEV was self-cleaved.

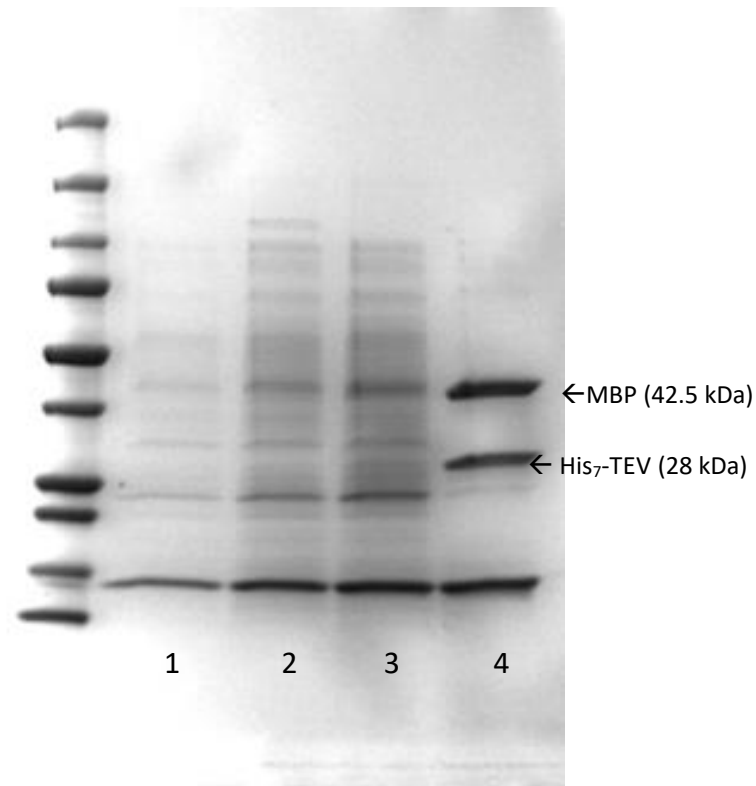


Fig. 1. SDS\_PAGE containing all four small culture flasks. The first lane represents cells without plasmid pMHT $\Delta$ 238 and were not induced with IPTG. The second lane represents cells without plasmid pMHT $\Delta$ 238 and were induced with IPTG. The third lane represents cells with plasmid pMHT $\Delta$ 238 but were not induced with IPTG. The last lane was cells with plasmid pMHT $\Delta$ 238 and were induced with IPTG.

### *Large Scale Expression*

Overnight growth media were prepared in 50mL of LB with kanamycin (50mg/mL). Colonies were inoculated from the transformation plates of pMHT $\Delta$ 238 into BL21(DE3) C+ RP cells. The overnight culture was incubated overnight at 37°C. A fraction of the overnight culture was added to 1L of the expression medium containing

LB and Kanamycin. When the OD<sub>600</sub> reached about 0.5, the culture was induced with 10 mL of IPTG. This was incubated for ~5-10 hours at 30°C. Cells were harvested by centrifugation at 6000rpm for 5 minutes. Cells were then lysed using the French press. Lysis buffer was 20mM sodium phosphate (pH 7.4), 500mM sodium chloride, and 20mM imidazole. Lysate was stored at 4°C overnight.

### *His7-TEV Purification*

Affinity chromatography was used to purify His7-TEV. Buffer A was 20mM sodium phosphate (pH 7.4), 500mM sodium chloride. Buffer B was 20mM sodium phosphate (pH 7.4), 500mM sodium chloride, 500mM imidazole. His Trap FF 5mL column was used and the preprogrammed method (AS His Trap FF 5ml) on IMAC ensured proper buffer exchange throughout the purification. The flow rate was set to 5mL/min. By increasing the concentration of imidazole, His7-TEV was eluted into fractions which was approximately 22mL all together.

Dialysis was used to desalt the eluted fractions. The equilibrium buffer was 20mM sodium phosphate (pH 7.4), 500mM sodium chloride. The membrane containing eluted fractions was submerged in equilibrium buffer on a stir plate overnight at 4°C. Approximately 18mL was recovered from overnight dialysis. For long-term storage, glycerol was added to get a final concentration of 20%. Samples were transferred into 1 mL microfuge tubes and stored in -80°C freezer.

### **Discussion**

I was able to successfully purify His7-TEV in the laboratory. I tested the activity of purified His7-TEV to make sure it can properly cleave target proteins. Cg10062

(100mM sodium phosphate, pH 8, 40% glycerol) was obtained from Amaya in Karen Draths Laboratory. I incubated the treatment tube with Cg10062 and His7-TEV, while the control tube only had Cg10062. By using affinity chromatography, I was able to look at the flow-through to see if His7-TEV worked properly. The control tube showed Cg10062 with the TEV site and His tag was eluted during the elution stage of the purification system. The treatment tube showed the target protein Cg10062 free from its histidine tag and was eluted in the flow-through. His7-TEV and His6-TEV site was eluted during the elution stage. This confirmed the activity of TEV protease, which successfully cleaved the target protein.

His7-TEV is now available to the members of Dr. Draths laboratory. This sequence specific protease can now be used to cleave proteins of their histidine tags. To yield more His7-TEV next time, the expression culture after induction with IPTG can stay longer in the 30°C shaking incubator. This will allow for more TEV to be produced because more cells will accumulate. This results in larger pellets which are then centrifuged.

### **References**

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